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Laparoscopic Hartmann's procedure: is it a good option for acute colonic diverticulitis with generalized peritonitis? A national multicenter comparative matched cohort study

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Cover Letter

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May 5th, 2026

Professor Justin B. Dimick,

Dear Editor-in-Chief of Annals of Surgery

Please find enclosed our manuscript entitled: "Laparoscopic Hartmann's procedure: is it a good option for acute colonic diverticulitis with generalized peritonitis? A national multicenter comparative matched cohort study" for consideration as an original investigation.

This manuscript is an original contribution; all authors have read and complied with author guidelines. Authors, status and affiliations are at the end of this letter.

Acute perforated sigmoid diverticulitis with generalized peritonitis remains a challenging surgical emergency. Although recent trials and current guidelines favor resection with primary anastomosis in stable patients, Hartmann's procedure is still used in more complex or high-risk situations. At the same time, minimally invasive colorectal surgery has progressively expanded to emergency settings, and laparoscopic Hartmann's procedure has been proposed as an attractive strategy. In theory, laparoscopy could reduce postoperative morbidity, and limit adhesion formation, thereby facilitating subsequent restoration of bowel continuity. This minimally invasive two-stage strategy could therefore appear to combine the immediate safety of a non-restorative procedure with improved long-term functional outcomes. However, whether these expected benefits are achieved in generalized diverticular peritonitis remains uncertain.

The aim of this multicenter national matched cohort study was to compare laparoscopic Hartmann's procedure with open sigmoidectomy and primary anastomosis with diverting ileostomy, with particular attention to postoperative outcomes and 12-month stoma-free survival. Our results indicate that laparoscopic Hartmann's procedure is feasible in this setting, but the anticipated advantages of the minimally invasive approach were not demonstrated.

Short-term outcomes were comparable between strategies, with no significant reduction in postoperative morbidity, major complications, mortality, or length of stay. In addition, conversion was frequent. Most importantly, laparoscopic Hartmann's procedure was associated with fewer stoma reversals and poorer 1-year stoma-free survival than open primary anastomosis with diversion.

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Thank you for receiving our manuscript and considering it for publication. We hope that you will find it interesting and suitable for publication in Annals of Surgery

Sincerely yours,

Antoine Mariani, M.D., PhD

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Mini-Abstract

In this French national propensity-matched cohort study of generalized diverticular peritonitis, laparoscopic Hartmann's procedure and open primary anastomosis with diverting ileostomy yielded similar short-term outcomes. However, laparoscopic Hartmann's procedure was associated with fewer stoma reversals and poorer 1-year stoma-free survival.

Abstract

Objective

Compare laparoscopic Hartmann's procedure (LHP) with open sigmoidectomy with primary anastomosis and diverting ileostomy (OPA).

Summary Background Data

Laparoscopic surgery is increasingly used in emergency settings, but its role in generalized diverticular peritonitis remains debated.

Methods

Retrospective study using a French multicenter cohort, including 7,053 patients who underwent surgery for colonic diverticulitis. Patients were included if they had a generalized peritonitis (Hinchey III or IV) treated in emergency by laparoscopic Hartmann or open anastomosis with ileostomy. Perioperative outcomes and stoma-free survival were compared after propensity score matching.

Results

284 patients met inclusion criteria (138 LHP, 146 OPA). After matching, 101 patients were included in each group. Baseline characteristics were comparable. 30% of the overall population had Hinchey IV diverticulitis. Conversion was required in 49% of LHP cases. LHP did not confer significant benefits concerning postoperative outcomes including 90-day morbidity (53% vs 50%, $p = 0.8$), major complications (27% vs 26%, $p > 0.9$), 90-day mortality (5% vs 3%, $p = 0.7$), or length of hospital stay (12 vs 10 days, $p = 0.095$), and was associated with fewer stoma reversal (70% vs 88%, $p = 0.002$). Stoma-free survival at 1 year was significantly better in the OPA group (85% vs 65%, respectively; log-rank $p < 0.0001$; Cox model HR = 1.84 [1.39–2.44], $p < 0.001$). No treatment-by-Hinchey interaction was observed ($p = 0.269$).

Conclusion

Although early outcomes were comparable, the minimally invasive Hartmann's procedure was associated with fewer stoma reversals and poorer stoma-free survival. When feasible, open primary anastomosis should remain the preferred approach.

MeSH Keywords :

Diverticulitis; Laparoscopy; Perforated diverticulitis; Sigmoid resection, Hartmann Procedure, Propensity score.

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Laparoscopic Hartmann's procedure: is it a good option for acute colonic diverticulitis with generalized peritonitis? A national multicenter comparative matched cohort study

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Conference presentation

The results reported in this paper were presented as a poster presentation at the 20th congress of ESCP (September 6th, 2025) and oral communication at the 1st mutual French annual congress of surgery (3CVD). The manuscript, including related data, figures and tables has not been previously published and the manuscript is not under consideration elsewhere.

Contribution of each author:

All authors confirm that they meet the current criteria of the International Committee of Medical Journal Editors (ICMJE) for authorship.

All authors did:

- Substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data for the work;
- And drafted the work or reviewed it critically for important intellectual content
- Approved this version to be published
- And agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

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Text of the Paper

Introduction

Acute perforated sigmoid diverticulitis with generalized peritonitis (Hinchey III–IV) is a surgical emergency requiring rapid source control while balancing postoperative morbidity and the risk of a permanent stoma (1,2). Historically, Hartmann's procedure was the standard of care but was associated with substantial morbidity, non-negligible mortality, and low reversal rates (3,4). Following recent randomized trials and meta-analyses, national and international guidelines have shifted the paradigm, now favoring sigmoid resection with primary anastomosis when possible, as it offers similar short-term safety but significantly higher reversal rates and better long-term stoma-free survival than Hartmann's Procedure, reserving the latter for unstable or very high-risk patients (5–12). Regarding the omission of a diverting stoma, guidelines remain cautious, suggesting unprotected anastomosis only for highly selected cases (13).

Over the same period, minimally invasive colorectal surgery has expanded into the emergency setting (14,15). While laparoscopic peritoneal lavage has fallen out of favor due to high rates of persistent sepsis, laparoscopic sigmoid resection has emerged as a feasible option(16–18). Current data suggest that in experienced hands, emergency laparoscopic resection could reduce parietal morbidity and hospital stay compared to open surgery (19–22). However, the specific role of the Laparoscopic Hartmann's Procedure (LHP) remains debated. Retrospective or registry-based studies suggest that this procedure can be performed safely and may reduce overall complications and length of stay compared with open Hartmann's procedure.

Crucially, the potential benefits of LHP may extend beyond the acute phase. The low reversal rate of the Hartmann's strategy is often attributed to dense intra-abdominal adhesions formed after open surgery. Emerging evidence suggests that performing the initial Hartmann's procedure laparoscopically may reduce adhesion formation, thereby facilitating a subsequent laparoscopic reversal (22,23). This "minimally invasive two-stage strategy" could potentially combine the immediate safety of a non-restorative procedure with a higher likelihood of successful future restoration of bowel continuity. It remains unclear, however, whether this minimally invasive approach can compete with the current standard of care. The aim of the present study was to evaluate, in a large national cohort, whether Laparoscopic Hartmann's Procedure offers comparable short-term outcomes and stoma-free survival to Open Sigmoid Resection with Primary Anastomosis and diverting ileostomy in patients admitted for generalized peritonitis (Hinchey III–IV) related to acute perforated sigmoid diverticulitis.

Methods

Patients

Data were obtained from a retrospective multicenter database including all adult patients operated on for sigmoid diverticulitis (SD) between January 2010 and September 2021 across 43 French centers. This analysis is part of a study registered at ClinicalTrials.gov (NCT06200857) (24). The need for informed consent was waived owing to the retrospective study design.

Patients were included if they underwent emergency surgery for SD peritonitis classified as Hinchey III or IV, treated either by open resection with primary anastomosis and diverting stoma or by laparoscopic Hartmann's procedure. Hinchey stage was determined from operative reports according to the original Hinchey classification and current guideline definitions, with stage III defined as generalized purulent peritonitis and stage IV as generalized fecal peritonitis (1,25).

Exclusion criteria were non-urgent surgery, Hinchey stages < III, surgical approaches other than the two evaluated strategies, American Society of Anesthesiologists (ASA) score = 4 and/or preoperative admission to an intensive care unit. These latter criteria were used as a surrogate for hemodynamic instability, a parameter not recorded in the database.

Endpoints

Primary Endpoint

The primary outcome was the 12-month stoma-free survival, defined as the time from the index emergency surgery to the successful restoration of intestinal continuity, with patients censored at death, loss to follow-up, or at 12 months if no stoma reversal had occurred.

Secondary Endpoints

Secondary outcomes included intraoperative morbidity (any adverse event occurring during the index procedure), the conversion rate in the LHP group, 90-day postoperative morbidity according to Clavien–Dindo classification(26), length of hospital stay, recurrent diverticulitis during follow-up, and the occurrence of incisional hernias.

Surgical procedure

Two types of procedure were assessed: the Laparoscopic Hartmann's Procedure (LHP group)(27), consisting of laparoscopic sigmoidectomy with rectal closure and an end colostomy, with or without conversion to open surgery; and the Open sigmoidectomy with a primary anastomosis and a defunctioning stoma (OPA group). In both groups, decision concerning the lowering of the splenic flexure and abdominal drainage were left at the surgeon's discretion.

Statistical analysis

In this observational study, propensity score matching was used to reduce selection bias related to the surgical approach and other confounding factors, ensuring comparability between groups by adjusting for baseline differences between patients undergoing either LHP or OPA. Propensity scores were estimated using multivariable logistic regression including all pre-specified covariates. These covariates comprised age, sex, smoking history, BMI, ASA score, immunodepression, anticoagulant therapy, previous laparotomy, and the Hinchey classification. These factors were selected as known confounders of both the choice of surgical approach and perioperative outcomes. An "optimal" case-control match without replacement was used, minimizing absolute differences in propensity scores (28). Covariate balance after matching was assessed using standardized mean differences (SMD), with values < 0.1 indicating adequate balance (Supplementary Figure S1) (28). Missing data were addressed using pairwise deletion. Continuous variables were reported as median ± interquartile range (IQR) and compared using paired t-tests for approximately normally distributed variables (normality assessed visually and with the Shapiro–Wilk test), or Wilcoxon signed-rank tests otherwise. Categorical variables were expressed as frequency (%) and compared within matched pairs using McNemar's test.

A Cox proportional hazards model was used to compare 12-month stoma-free survival between groups in the matched cohort. To account for the matched design and within-pair correlation, a robust sandwich standard errors clustered by matched pairs was applied (29). A treatment-by-Hinchey interaction was tested in the matched Cox model. Additionally, a supplementary analysis separated patients who underwent a laparoscopic Hartmann procedure with or without conversion, to explore whether conversion modified stoma-free survival.

Results are presented as hazard ratios (HR) with 95% confidence intervals (CI). Statistical analyses were performed using RStudio software version 4.5.1 (R Foundation for Statistical Computing, Vienna, Austria). A p-value < 0.05 was considered statistically significant.

Results

Preoperative characteristics

A total of 1,521 patients underwent surgery for diverticular peritonitis (Hinchey III or IV) between January 2010 and September 2021 across 43 centers, including 34 university hospitals. The final study population comprised 284 patients: 138 in the LHP group and 146 in the OPA group (Fig. 1). After propensity score matching, two comparable groups of 101 patients each were obtained, with adequate balance across all baseline covariates (SMD < 0.1 for covariates included; Supplementary Fig. S1).

Before matching, patients in the LHP group were older (median age 63 vs 50 years, p = 0.03), more frequently classified as ASA 3 (26% vs 17%, p = 0.04), and had a higher proportion of Hinchey IV peritonitis (37% vs 23%, p = 0.01). Other baseline characteristics were similar between groups.

After matching, severe diverticulitis remained frequent, with Hinchey IV peritonitis observed in 31% of patients in the LHP group and 29% in the OPA group, and no other differences in preoperative characteristics were observed (Table 1).

Intraoperative findings and adverse events

In the propensity score–matched (PSM) cohort, intraoperative outcomes are summarized in Table 2. Median operative time tended to be shorter in the LHP group compared with the OPA group (138 [100–169] vs 180 [120–240] minutes), although this difference did not reach statistical significance (p = 0.065). Conversion to open surgery occurred in 49% of patients undergoing LHP.

Splenic flexure mobilization was performed far less frequently in the LHP group than in the OPA group (15% vs 72%, p < 0.001), as was high ligation of the inferior mesenteric artery (4.9% vs 14%, p = 0.003), with high ligation defined as division of the inferior mesenteric artery proximal to the origin of the left colic artery (30). As expected, anastomosis was performed exclusively in the OPA group, most commonly as a latero-lateral configuration (59%), Intraoperative transfusion rates were low and comparable between groups (3.3% vs 4.3%, p > 0.9).

Postoperative outcomes

Postoperative outcomes are detailed in Table 3. Overall 90-day morbidity was high but comparable between groups (53% in the LHP group vs 50% in the OPA group, p = 0.6). Ninety-day mortality did not significantly differ (5.0% vs 3.0%, p = 0.5), nor did the rate of intensive care unit admission (16% vs 11%, p = 0.3).

The severity of postoperative complications was similar between groups, with no difference in the distribution of Clavien–Dindo grades (p = 0.5). Severe complications (Clavien–Dindo >2) occurred in 27% of patients in the LHP group and 26% in the OPA group (p > 0.9). Anastomotic leakage was observed in 8 patients (8%) in the OPA group. Reintervention rates for complication were comparable between two groups (10% vs 16%, p = 0.2), as well as median length of stay and thirty-day readmission rates were comparable.

Longer-term postoperative events, including incisional hernia (21% vs 23%), and recurrence of diverticulitis, were not significantly different between groups.

Anastomotic stenosis concerned 2 patients (2,2%) in OPA group.

Stoma-Free Survival

At 12 months, the probability of being stoma-free was higher in the OPA group than in the LHP group (85% vs 65%, respectively; stratified log-rank p < 0.0001) (Figure 1).

The median follow-up duration, estimated using the reverse Kaplan-Meier method, was 29 months in the LHP group and 70 months in the OPA group. In the Cox proportional hazards model accounting for the matched design with robust standard errors, OPA was associated with a significantly higher rate of stoma closure compared with LHP (HR = 1.84, 95% CI 1.39–2.44; $p < 0.001$), corresponding to a shorter time to stoma-free status (Figure 2). As an effect-modification analysis by peritonitis severity, the hazard ratio for stoma closure (OPA vs LHP) was 2.07 (95% CI, 1.44–2.99) in Hinchey III and 1.44 (95% CI, 0.88–2.38) in Hinchey IV, with no significant treatment-by-Hinchey interaction ($p = 0.269$).

Additionally, a supplementary analysis according to conversion status within the LHP group was performed (Supplementary Fig. S2). OPA was associated with a higher rate of stoma closure compared with both converted (HR = 1.59, 95% CI 1.12–2.27; $p = 0.01$) and non-converted LHP (HR = 2.09, 95% CI 1.44–3.01; $p < 0.001$). No significant difference in stoma-free survival was observed between converted and non-converted LHP (HR = 1.34, 95% CI 0.85–2.13; $p = 0.21$).

Discussion

Randomized trials have established primary anastomosis as the preferred reconstructive strategy over Hartmann's procedure for perforated diverticulitis with generalized peritonitis. In both the LADIES (DIVA arm) and DIVERTI trials, short-term outcomes were broadly comparable between strategies, whereas long-term stoma outcomes favored primary anastomosis (8,10). However, these trials were not designed to assess surgical approach and do not address whether laparoscopy can mitigate the long-term limitations of a non-restorative strategy. Our study aimed to specifically study the impact of laparoscopy when performing a Hartmann's procedure (LHP) in comparison with the reference intervention including open sigmoid resection with primary anastomosis (OPA) and diverting ileostomy.

The main finding of this propensity score–matched comparison is that long-term stoma-free survival was lower after LHP than after OPA, despite similar short-term outcomes. At 12 months, the probability of being stoma-free was 85% in the OPA group versus 65% in the LHP group (log-rank $p < 0.0001$), with an almost twofold higher hazard of stoma closure after OPA than after LHP (HR 1.84, 95% CI 1.39-2.44). Importantly, we found no statistically significant interaction between treatment strategy and Hinchey stage, suggesting that the long-term stoma-free survival advantage associated with OPA was not restricted to one severity stratum. This interaction analysis should nonetheless be interpreted cautiously given the lower precision in Hinchey IV.

Accordingly, we selected open primary anastomosis with diverting ileostomy as the reference strategy, as it reflects contemporary practice for stable patients with generalized peritonitis. Although omission of diversion has been proposed in selected cases, current guidelines remain cautious and generally restrict unprotected anastomosis to highly selected patients.(5–7) The anastomotic leak rate observed in the OPA group (8%) in our study warrants specific discussion, as it directly intersects with the issue of diversion. A protective stoma can effectively mitigate the clinical consequences of an anastomotic leak; however, it is also associated with its own morbidity, and its reversal requires an additional operation, which has been reported to carry complication rates at 14 – 21% (31,32). Generalized peritonitis has historically raised concerns regarding anastomotic safety. Nevertheless, contemporary randomized trials consistently demonstrate acceptable leak rates even in the absence of diversion, with short-term outcomes comparable to those of primary anastomosis with diverting ileostomy (LADIES and DIVERTI trials). The present results do not provide sufficient evidence to advocate routine omission of diversion in this setting. Rather, they support a pragmatic interpretation whereby a protected anastomosis represents a reasonable compromise between short-term safety and the long-term benefit of higher stoma-free survival (13,33,34). The randomized controlled trial DIVERTI2 specifically investigated the role of a protective stoma following primary anastomosis for Hinchey III-IV peritonitis patients, with overall morbidity at one year as the primary endpoint (13). To avoid stoma formation, the latest alternative remains laparoscopic lavage. Based principally on three randomized controlled trials (DILALA, LOLA, and SCANDIV), this intervention has been shown to be safe and technically straightforward; however, it is associated with notable drawbacks, including a higher risk of recurrent diverticulitis and an increased need for reinterventions.

The present findings can be positioned within the limited and heterogeneous observational literature on minimally invasive emergency surgery for perforated diverticulitis. Vennix et al. reported that emergency laparoscopic sigmoid resection is feasible in selected settings and may be associated with lower morbidity and shorter hospital stay compared with open surgery, but their analysis primarily addressed the impact of surgical approach within mixed reconstructive strategies and therefore cannot establish strategy-level equivalence (20). Turley et al., using a large propensity-matched NSQIP analysis restricted to Hartmann's procedures, found no significant short-term advantage of laparoscopy over open surgery, although interpretation is limited by the absence of granular clinical data and by the lack of information on Hinchey status or on long-term stoma outcomes (19). At the other end of the spectrum, Cassini et al. described a totally laparoscopic two-step strategy with high feasibility of laparoscopic reversal in selected patients treated in expert centers, providing proof of technical feasibility rather than comparative evidence against reconstructive strategies (22). Taken together, these studies support that LHP is feasible and may improve selected perioperative metrics in some contexts, but they do not demonstrate equivalence with the contemporary benchmark strategy of primary anastomosis with diversion for long-term stoma-free survival, an issue directly addressed by the present matched comparative design

A key complementary result comes from the analysis according to conversion status within the LHP group, which occurred in approximately half of cases. This is similar to Vennix et al. (≈54%) and contrasts with the lower rate reported in a highly selected expert series by Cassini et al. (10%) (20,22). Conversion rates cannot be reliably extracted from registry-based analyses such as Turley et al. (19,20,22). Despite this high conversion rate, OPA remained associated with a higher rate of stoma closure compared with both converted and non-converted LHP, while no significant difference in stoma-free survival was observed between converted and non-converted LHP. These findings indicate that the unfavorable long-term trajectory after LHP is not explained by secondary conversion to an open Hartmann's procedure but rather reflects the intrinsic limitations of the Hartmann strategy itself, regardless of surgical approach. The present dataset did not allow analysis of the specific causes of conversion, which prevents exploring whether conversion was driven by disease severity, technical difficulty, or intraoperative findings.

Laparoscopic Hartmann's procedure has been advocated on theoretical grounds in this emergency setting, including reduced parietal trauma and wound morbidity, potentially faster postoperative recovery, and the hypothesis that a minimally invasive first stage may facilitate subsequent reversal by limiting adhesions (35). However, in our study, short-term outcomes were similar between strategies, with no significant differences in 90-day morbidity or mortality. This aligns with findings from Turley et al., in which a laparoscopic Hartmann's procedure did not significantly reduce postoperative morbidity or mortality compared with open surgery in emergency diverticulitis, with comparable 30-day mortality (4.5% open vs 3.0% laparoscopic) and overall morbidity (30% vs 25%) between matched groups (19). Although laparoscopy is associated with improved outcomes in elective colorectal surgery, recent reviews suggests these benefits are not consistently seen in emergency settings, and reviews of emergency laparoscopic colorectal surgery indicate that the technical challenges of peritonitis can limit the expected advantages of a minimally invasive approach (33,34,36,37).

This study has several strengths that support the robustness and relevance of the present findings. It relies on a national multicenter surgical database (AFC), provides one of the largest reported series specifically focused on laparoscopic Hartmann's procedure in this emergency indication, and uses propensity score matching to reduce baseline imbalance when benchmarking LHP against the contemporary reference strategy. The present findings can be positioned within the limited and heterogeneous observational literature. As a retrospective analysis, the comparison remains exposed to residual confounding despite propensity-score matching. Key clinical variables were not available, including the initial intention regarding restoration of continuity, which may have influenced both treatment allocation and subsequent stoma management. Approximately half of LHP required conversion, and the dataset did not capture the reasons for conversion. Additionally, detailed information on the modalities of restoration of continuity was not available (surgical approach at reversal and whether reconstruction was performed with or without diversion), and quality-of-life outcomes were not collected.

Conclusions

Our findings support a pragmatic interpretation for current practice in generalized diverticular peritonitis. In this matched national cohort, performing Hartmann's procedure laparoscopically did not translate into clearly improved short-term outcomes and did not offset the long-term functional disadvantage inherent to a non-restorative strategy. When patients are stable and eligible for reconstruction, open sigmoid resection with protected primary anastomosis therefore remains the reference approach to maximize the likelihood of long-term stoma-free survival whereas Hartmann's procedure should be reserved for unstable or very high-risk patients.

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Legend for Illustrations

Figure 1: Flow chart for patient inclusion

Flow diagram of patient selection and propensity score matching. Among 1521 patients in the database, 284 met inclusion criteria (LHP n = 138; OPA n = 146). After 1:1 propensity score matching, 101 patients were included in each group.

Table 1: Preoperative characteristics

Table 2: Intraoperative outcomes of propensity-score matched patients

Table 3: Postoperative outcomes of propensity-score matched patients

Figure 2: Stoma free survival according to surgical approach

Kaplan–Meier estimates of 12-month stoma-free survival in propensity score–matched patients undergoing LHP (red) versus OPA (blue). Comparisons used a stratified log-rank test; HRs were derived from a Cox model with robust variance accounting for matched pairs. Numbers at risk are shown.

List of Supplemental Digital Content

Supplementary Figure S1: Covariate balance before and after propensity score matching.

Absolute standardized mean differences (SMD) for all covariates in the unmatched and matched samples. The vertical dashed line indicates the predefined threshold for acceptable balance (SMD = 0.1). All covariates achieved SMD < 0.1 after matching, , indicating adequate covariate balance between groups.

Supplementary Figure S2: Stoma-free survival according to surgical approach and conversion status

Kaplan–Meier estimates of stoma-free survival up to 12 months after index surgery in propensity score–matched patients, comparing laparoscopic Hartmann’s procedure not converted to open surgery (LHP non-converted, orange), LHP converted to open surgery (LHP converted, pink), and open procedure (OPA, blue). Follow-up was censored at 12 months. Hazard ratios and corresponding p-values were estimated using Cox proportional hazards models with robust standard errors clustered by matched pairs. Numbers at risk at selected time points are displayed below the plot.

Table 1: Preoperative characteristics

	All patients (n=284)			Matched patients (n=202)		
Characteristic	**LHP (n=138)**	**OPA (n=146)**	**p**	**LHP (n=101)**	**OPA (n=101)**	**p**
Age (year)	63 [51, 76]	50 [50, 69]	0.03*	58 [49, 72]	60 [51, 73]	0.2
Gender, male	73 (52,9%)	88 (59,9%)	0.2	58 (57%)	55 (55%)	0.6
ASA score > 2	36 (26%)	25 (17%)	0.04*	21 (21%)	18 (18%)	0.6
BMI kg/m2	25.5 [22.3, 28.3]	25.4 [22.9, 29.4]	0.7	25.5 [22.3, 28.3]	25.1 [21.7, 28.6]	0.9
Active smoker	30 (21,7%)	45 (30,6%)	0.09	30 (30%)	27 (27%)	0.6
Immunodepression	24 (17,4%)	14 (9,5%)	0.051	15 (15%)	14 (14%)	0.8
Anticoagulation	13 (9,4%)	9 (6,1%)	0,3	8 (8.5%)	5 (5.1%)	0.2
Medical antecedents						
Cardiologic	50 (36,2%)	59 (40,1%)	0.5	29 (30%)	41 (42%)	0.13
Neurologic	23 (16,7%)	19 (12,9%)	0.4	14 (14%)	12 (12%)	0.7
Pulmonary	24 (17,4%)	18 (12,2%)	0.2	15 (16%)	11 (11%)	0.6
Diabetes	10 (7,2%)	10 (6,8%)	0.9	6 (6.2%)	7 (7.1%)	0.6
Previous laparotomy	15 (10,9%)	13 (8,8%)	0.6	10 (9.9%)	12 (12%)	0.6
Previous diverticulitis	29 (21%)	41 (27,9%)	0.2	23 (24%)	28 (29%)	0.5
Actual diverticulitis						
Hinchey Score						
III	87 (63%)	113 (76,9%)	0.01*	70 (69%)	72 (71%)	0.7
IV	51 (37%)	34 (23,1%)	0.01*	31 (31%)	29 (29%)	0.7
Associated hemorrhage	1 (0,7%)	0 (0%)	0.5	1 (1.0%)	0 (0%)	
Associated occlusion	7 (5,1%)	10 (6,8%)	0.5	6 (5.9%)	6 (5.9%)	>0.9
Associated fistula	2 (1,4%)	5 (3,4%)	0.4	2 (2.0%)	4 (4.0%)	0.4
Surgery within 24 hours of diagnosis	134 (97,1%)	139 (94,6%)	0.3	98 (97%)	98 (97%)	>0.9
Surgery after 24 hours	1 (0,7%)	5 (3,4%)	0.6	1 (1.0%)	1 (1.0%)	>0.9

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

ASA, American Society of Anesthesiologists; BMI, body mass index; CRP, C-reactive protein;SD, standard deviation.

*Statistically significant differences.

Table 2: Intraoperative outcomes of propensity-score matched patients

Intraoperative outcome	LHP (N = 101)	OPA (N = 101)	p-value
Operative time, min*	138 [100, 169]	180 [120, 240]	0.065
Conversion to open surgery†	49 (49%)	Not concerned	
Splenic flexure mobilization*	15 (15%)	73 (72%)	<0.001*
High IMA ligation*	5 (4.9%)	14 (14%)	0.003*
Anastomose type			
Latero-terminal	Not concerned	60 (59%)	
Termino-terminal	Not concerned	32 (32%)	
Transfusion	3 (3.3%)	4 (4.3%)	>0.9

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

IMA, Inferior Mesentery Artery

* Statistically significant differences.
† Reasons for conversion are unknown as they were not collected in the database.

Table 3: Postoperative outcomes of propensity-score matched patients

Postoperative outcome	LHP (N = 101)	OPA (N = 101)	p-value
Overall 90 days morbidity	54 (53%)	50 (50%)	0.6
Overall 90 days mortality	5 (5.0%)	3 (3.0%)	0.5
ICU hospitalization, %	16 (16%)	11 (11%)	0.3
Dindo-Clavien			0.5
1	29 (29%)	34 (34%)	
2	24 (24%)	19 (19%)	
3	8 (7.9%)	13 (13%)	
4	14 (14%)	10 (9.9%)	
5	5 (5.0%)	3 (3.0%)	
Dindo-Clavien >2	27 (27%)	26 (26%)	>0.9
Intraperitoneal abscess	7 (7.1%)	9 (9.0%)	0.6
Anastomotic leakage		8 (8.0%)	
Length of hospital stay, d	12 [8, 19]	10 [8, 16]	0.095
30-day readmission	15 (16%)	19 (21%)	0.5

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

ICU, intensive care unit; NS, not significant

Figure 1: Flow chart for patient inclusion

Flow diagram of patient selection and propensity score matching. Among 1521 patients in the database, 284 met inclusion criteria (LHP n = 138; OPA n = 146). After 1:1 propensity score matching, 101 patients were included in each group.

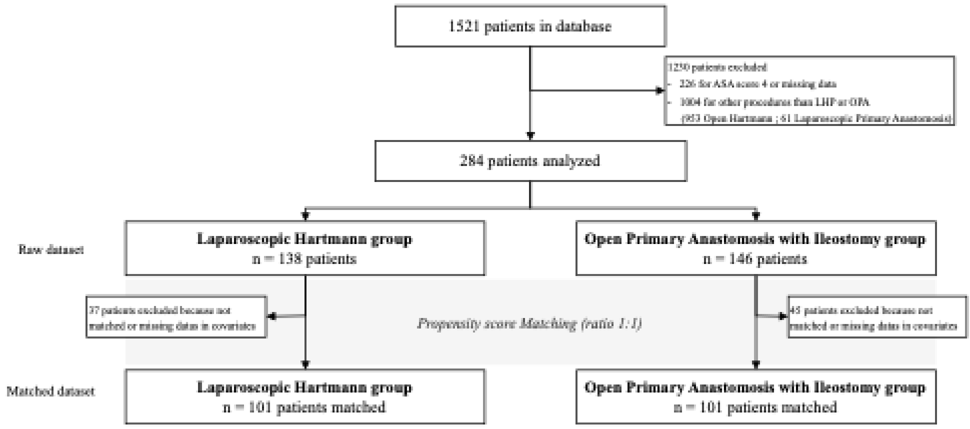
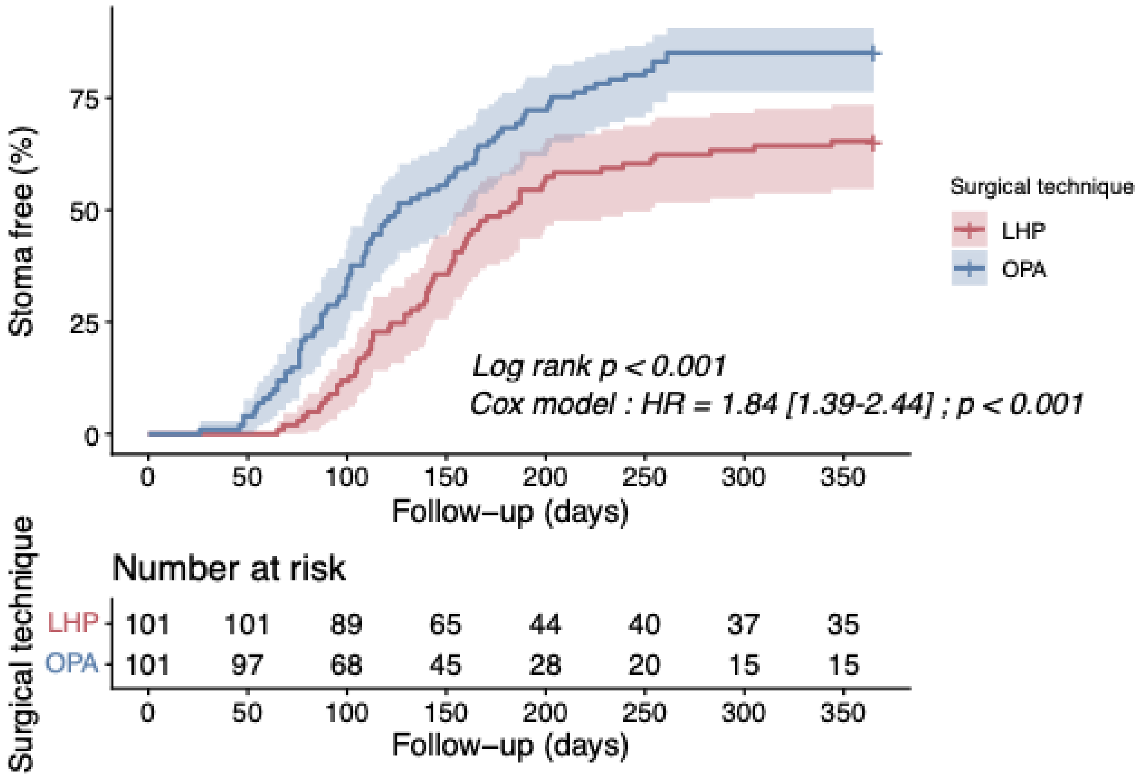


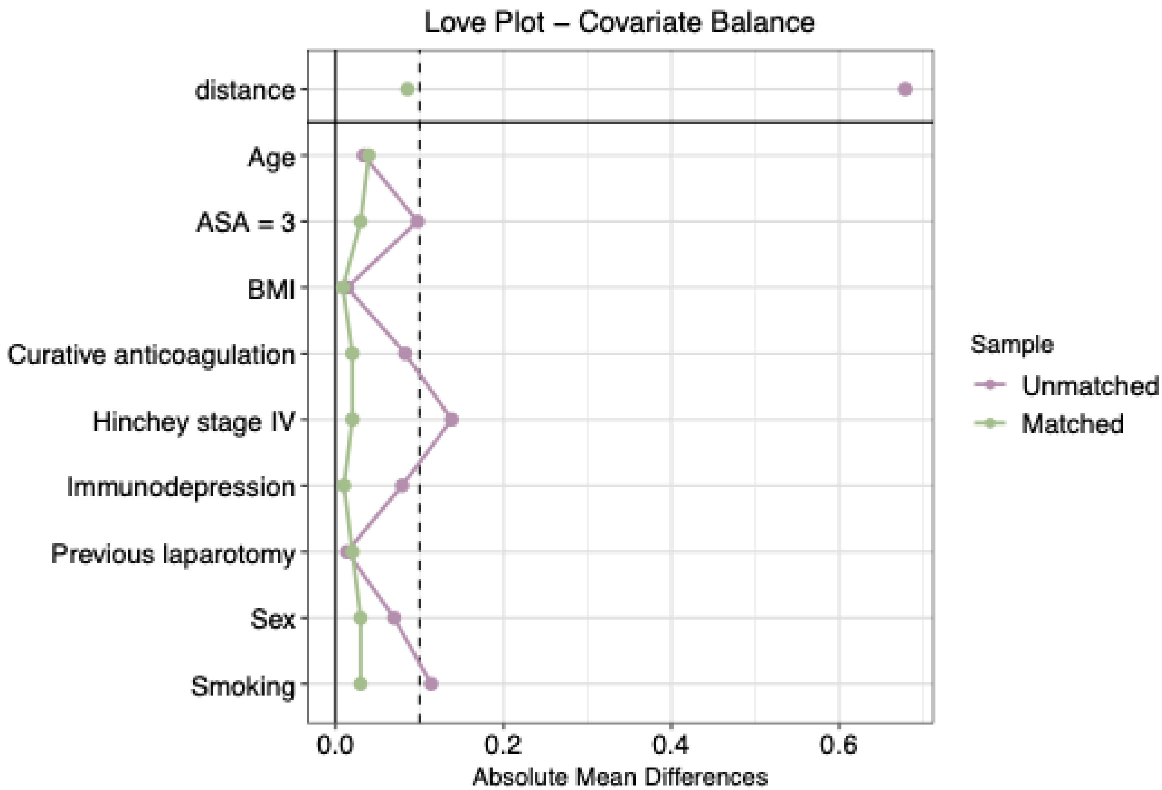
Figure 2: Stoma free survival according to surgical approach

Kaplan–Meier estimates of 12-month stoma-free survival in propensity score–matched patients undergoing LHP (red) versus OPA (blue). Comparisons used a stratified log-rank test; HRs were derived from a Cox model with robust variance accounting for matched pairs. Numbers at risk are shown.



Supplementary Figure S1: Covariate balance before and after propensity score matching.

Absolute standardized mean differences (SMD) for all covariates in the unmatched and matched samples. The vertical dashed line indicates the predefined threshold for acceptable balance (SMD = 0.1). All covariates achieved SMD < 0.1 after matching, , indicating adequate covariate balance between groups.



Supplementary Figure S2: Stoma-free survival according to surgical approach and conversion status

Kaplan–Meier estimates of stoma-free survival up to 12 months after index surgery in propensity score–matched patients, comparing laparoscopic Hartmann's procedure not converted to open surgery (LHP non-converted, orange), LHP converted to open surgery (LHP converted, pink), and open procedure (OPA, blue). Follow-up was censored at 12 months. Hazard ratios and corresponding p-values were estimated using Cox proportional hazards models with robust standard errors clustered by matched pairs. Numbers at risk at selected time points are displayed below the plot.

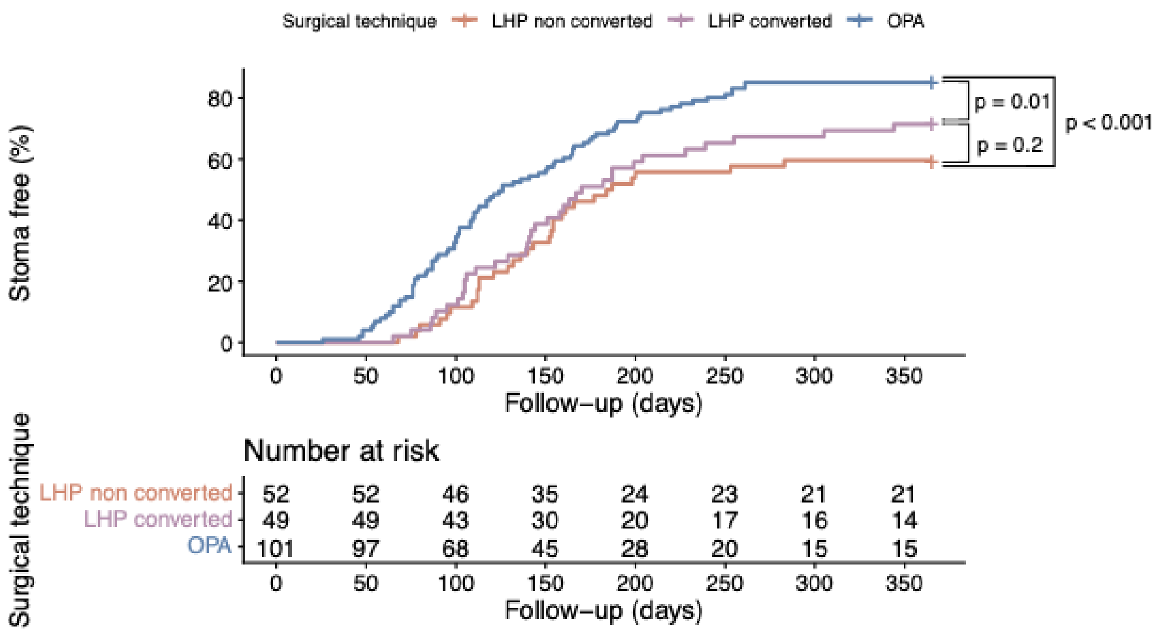


Table 1 : Patient characteristics

Table 1: Preoperative characteristics

	All patients (n=284)			Matched patients (n=202)		
Characteristic	**LHP (n=138)**	**OPA (n=146)**	**p**	**LHP (n=101)**	**OPA (n=101)**	**p**
Age (year)	63 [51, 76]	50 [50, 69]	0.03*	58 [49, 72]	60 [51, 73]	0.2
Gender, male	73 (52,9%)	88 (59,9%)	0.2	58 (57%)	55 (55%)	0.6
ASA score > 2	36 (26%)	25 (17%)	0.04*	21 (21%)	18 (18%)	0.6
BMI kg/m2	25.5 [22.3, 28.3]	25.4 [22.9, 29.4]	0.7	25.5 [22.3, 28.3]	25.1 [21.7, 28.6]	0.9
Active smoker	30 (21,7%)	45 (30,6%)	0.09	30 (30%)	27 (27%)	0.6
Immunodepression	24 (17,4%)	14 (9,5%)	0.051	15 (15%)	14 (14%)	0.8
Anticoagulation	13 (9,4%)	9 (6,1%)	0,3	8 (8.5%)	5 (5.1%)	0.2
Medical antecedents						
Cardiologic	50 (36,2%)	59 (40,1%)	0.5	29 (30%)	41 (42%)	0.13
Neurologic	23 (16,7%)	19 (12,9%)	0.4	14 (14%)	12 (12%)	0.7
Pulmonary	24 (17,4%)	18 (12,2%)	0.2	15 (16%)	11 (11%)	0.6
Diabetes	10 (7,2%)	10 (6,8%)	0.9	6 (6.2%)	7 (7.1%)	0.6
Previous laparotomy	15 (10,9%)	13 (8,8%)	0.6	10 (9.9%)	12 (12%)	0.6
Previous diverticulitis	29 (21%)	41 (27,9%)	0.2	23 (24%)	28 (29%)	0.5
Actual diverticulitis						
Hinchey Score						
III	87 (63%)	113 (76,9%)	0.01*	70 (69%)	72 (71%)	0.7
IV	51 (37%)	34 (23,1%)	0.01*	31 (31%)	29 (29%)	0.7
Associated hemorrhage	1 (0,7%)	0 (0%)	0.5	1 (1.0%)	0 (0%)	
Associated occlusion	7 (5,1%)	10 (6,8%)	0.5	6 (5.9%)	6 (5.9%)	>0.9
Associated fistula	2 (1,4%)	5 (3,4%)	0.4	2 (2.0%)	4 (4.0%)	0.4
Surgery within 24 hours of diagnosis	134 (97,1%)	139 (94,6%)	0.3	98 (97%)	98 (97%)	>0.9
Surgery after 24 hours	1 (0,7%)	5 (3,4%)	0.6	1 (1.0%)	1 (1.0%)	>0.9

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

ASA, American Society of Anesthesiologists; BMI, body mass index; CRP, C-reactive protein; SD, standard deviation.

*Statistically significant differences.

Table 2 : Intraoperative results

Table 2: Intraoperative outcomes of propensity-score matched patients

Intraoperative outcome	**LHP (N = 101)**	**OPA (N = 101)**	**p-value**
Operative time, min*	138 [100, 169]	180 [120, 240]	0.065
Conversion to open surgery†	49 (49%)	Not concerned	
Splenic flexure mobilization*	15 (15%)	73 (72%)	<0.001*
High IMA ligation*	5 (4.9%)	14 (14%)	0.003*
Anastomose type			
Latero-terminal	Not concerned	60 (59%)	
Termino-terminal	Not concerned	32 (32%)	
Transfusion	3 (3.3%)	4 (4.3%)	>0.9

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

IMA, Inferior Mesentery Artery

* Statistically significant differences.
† Reasons for conversion are unknown as they were not collected in the database.

Table 3 : Postoperative results

Table 3: Postoperative outcomes of propensity-score matched patients

Postoperative outcome	LHP (N = 101)	OPA (N = 101)	p-value
Overall 90 days morbidity	54 (53%)	50 (50%)	0.6
Overall 90 days mortality	5 (5.0%)	3 (3.0%)	0.5
ICU hospitalization, %	16 (16%)	11 (11%)	0.3
Dindo-Clavien			0.5
1	29 (29%)	34 (34%)	
2	24 (24%)	19 (19%)	
3	8 (7.9%)	13 (13%)	
4	14 (14%)	10 (9.9%)	
5	5 (5.0%)	3 (3.0%)	
Dindo-Clavien >2	27 (27%)	26 (26%)	>0.9
Intraperitoneal abscess	7 (7.1%)	9 (9.0%)	0.6
Anastomotic leakage		8 (8.0%)	
Length of hospital stay, d	12 [8, 19]	10 [8, 16]	0.095
30-day readmission	15 (16%)	19 (21%)	0.5

Quantitative variables are expressed as median (interquartile range [IQR]), and qualitative variables are expressed as frequencies and percentages.

ICU, intensive care unit; NS, not significant

Figure 1 : Flow-chart

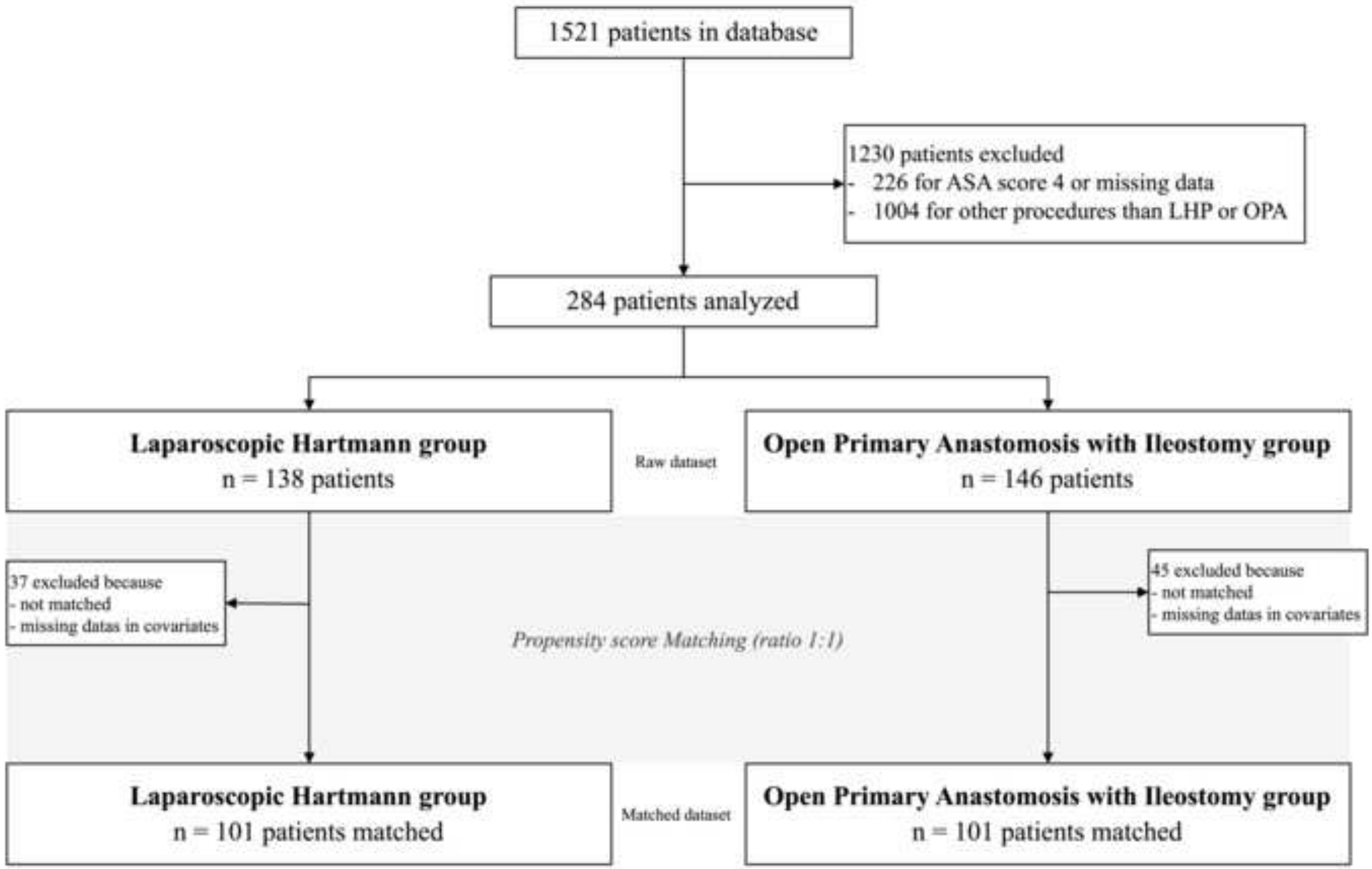


Figure 2 : Stoma-Free Survival

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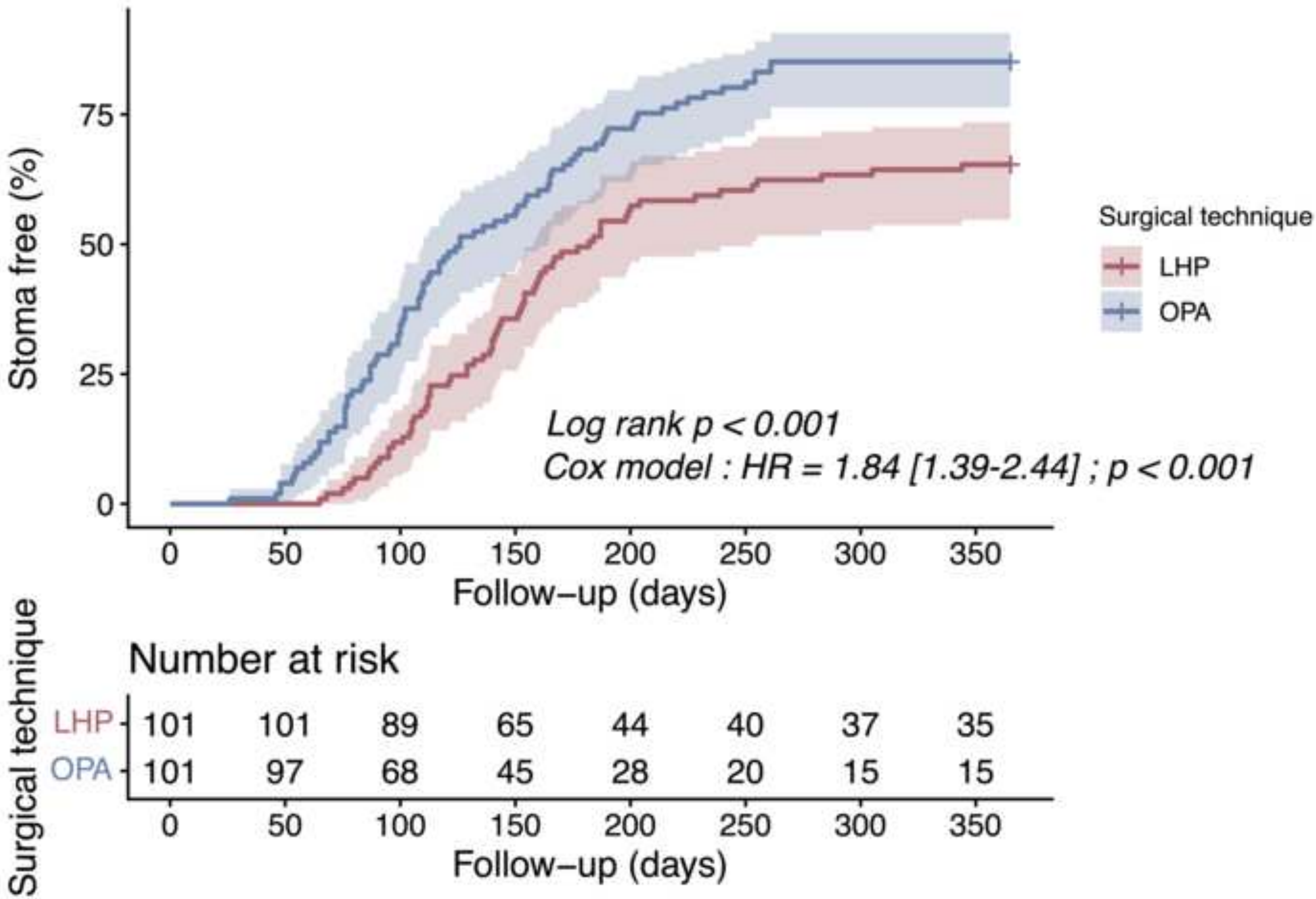

Conflict of interest/Disclosure

None of the authors have any conflict of interest.

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Supplemental Data File
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